

1(a)	(SI base unit of α =) $\text{m}^3 \text{s}^{-1} \times \text{m} / \text{m}^4 = \text{s}^{-1}$	A1
1(b)	(percentage uncertainty =) $3 + 4 + 2 \times 4 = 15$ (%)	A1
1(c)	$\alpha = QL / r^4$ $= 2.72 \times 10^{-8} \times 2.5 \times 10^{-2} / (7.1 \times 10^{-5})^4$ $= 2.7 \times 10^7$	C1
	absolute uncertainty = $0.15 \times [2.7 \times 10^7]$ $= 0.4 \times 10^7$	C1
	$\alpha = (2.7 \pm 0.4) \times 10^7 \text{s}^{-1}$	A1